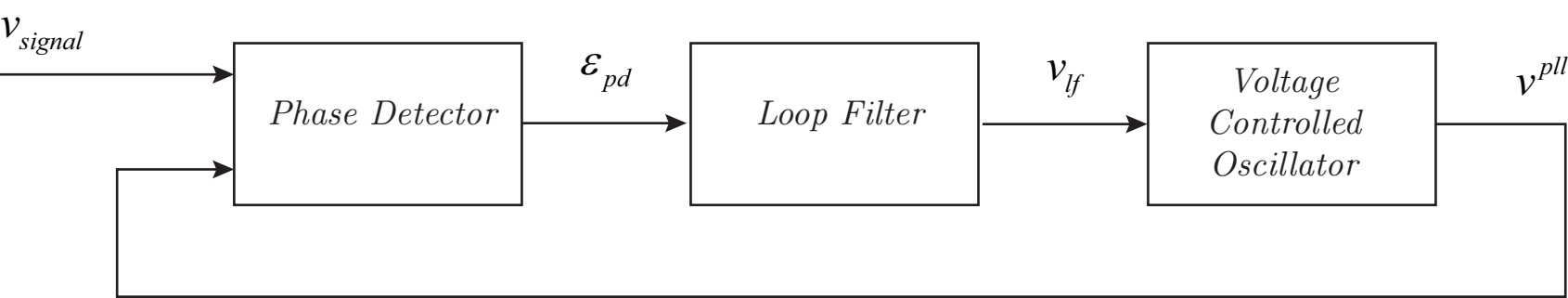
v_{signal}
Phase Detector
ε_{pd}
Loop Filter
v_{lf}
Voltage Controlled Oscillator
v^{pll}